

Corn Breeding Line Advancement -- Results Report

Precision Digital Agriculture Hackathon 2026 - Corn Breeding Track

Validated Results (real, held-out 2008 test)

Model	C1 corr / top-20%	C2 corr / top-20%
Within-population (family started)	0.636 / 51%	0.618 / 50%
Mate-selection (brand-new family)	0.317 / 40%	0.203 / 28%
Final deployed model (line-level)	0.167 / 29.5% (n=7429)	0.139 / 26.4% (n=8533)

All figures beat the 20% chance baseline for top-20% recovery. The deployed model combines mate-selection's family-level baseline (mid-parent genomic estimated breeding value, computed from a ridge regression trained on 2001-2007 data, alpha=1000) with each line's own within-family deviation, to produce a specific, ranked advancement recommendation for every genotyped 2008 line.

Cluster 1: Top 25 Recommended Lines to Advance

Full allocation: 300 lines across 64 families (300-plot budget, max 5 lines/family)

Rank	Line ID	Family	Pred. Yield Adv.	68% CI	Confidence
1	C1.398.15	C1.398	21.96	[11.0, 32.9]	Medium
2	C1.398.110	C1.398	21.59	[10.7, 32.5]	Medium
3	C1.398.121	C1.398	21.58	[10.7, 32.5]	Medium
4	C1.398.44	C1.398	21.31	[10.4, 32.2]	Medium
5	C1.398.103	C1.398	20.83	[9.9, 31.7]	Medium
6	C1.427.28	C1.427	19.21	[8.3, 30.1]	Medium
7	C1.427.36	C1.427	17.74	[6.8, 28.7]	Medium
8	C1.380.82	C1.380	17.0	[6.1, 27.9]	Medium
9	C1.382.76	C1.382	16.83	[5.9, 27.7]	Medium
10	C1.384.147	C1.384	16.39	[5.5, 27.3]	Medium
11	C1.442.13	C1.442	16.25	[5.3, 27.2]	Medium
12	C1.427.27	C1.427	16.21	[5.3, 27.1]	Medium
13	C1.427.11	C1.427	16.16	[5.2, 27.1]	Medium
14	C1.442.46	C1.442	16.1	[5.2, 27.0]	Medium
15	C1.427.19	C1.427	15.39	[4.5, 26.3]	Medium
16	C1.442.76	C1.442	15.17	[4.3, 26.1]	Medium
17	C1.384.19	C1.384	15.08	[4.2, 26.0]	Medium
18	C1.396.100	C1.396	14.89	[4.0, 25.8]	Medium
19	C1.380.160	C1.380	14.85	[3.9, 25.8]	Medium
20	C1.384.23	C1.384	14.84	[3.9, 25.8]	Medium
21	C1.442.86	C1.442	14.8	[3.9, 25.7]	Medium
22	C1.380.154	C1.380	14.59	[3.7, 25.5]	Medium
23	C1.380.137	C1.380	14.55	[3.6, 25.5]	Medium
24	C1.380.36	C1.380	14.48	[3.6, 25.4]	Medium
25	C1.439.5	C1.439	14.28	[3.4, 25.2]	Medium

Cluster 2: Top 25 Recommended Lines to Advance

Full allocation: 300 lines across 69 families (300-plot budget, max 5 lines/family)

Rank	Line ID	Family	Pred. Yield Adv.	68% CI	Confidence
1	C2.424.142	C2.424	16.11	[5.1, 27.1]	High
2	C2.375.142	C2.375	15.65	[4.6, 26.7]	Medium
3	C2.375.120	C2.375	15.58	[4.6, 26.6]	Medium
4	C2.406.32	C2.406	14.72	[3.7, 25.7]	High
5	C2.442.46	C2.442	14.23	[3.2, 25.2]	Low
6	C2.381.25	C2.381	14.2	[3.2, 25.2]	Medium
7	C2.435.153	C2.435	14.18	[3.2, 25.2]	Medium
8	C2.375.87	C2.375	13.97	[2.9, 25.0]	Medium
9	C2.381.49	C2.381	13.84	[2.8, 24.9]	Medium
10	C2.381.6	C2.381	13.8	[2.8, 24.8]	Medium
11	C2.375.187	C2.375	13.74	[2.7, 24.8]	Medium
12	C2.383.69	C2.383	13.73	[2.7, 24.8]	Medium
13	C2.381.15	C2.381	13.65	[2.6, 24.7]	Medium
14	C2.388.65	C2.388	13.51	[2.5, 24.5]	Medium
15	C2.406.79	C2.406	13.47	[2.5, 24.5]	High
16	C2.375.106	C2.375	13.45	[2.4, 24.5]	Medium
17	C2.381.19	C2.381	13.44	[2.4, 24.5]	Medium
18	C2.388.67	C2.388	13.23	[2.2, 24.3]	Medium
19	C2.411.83	C2.411	13.23	[2.2, 24.2]	Medium
20	C2.383.44	C2.383	12.95	[1.9, 24.0]	Medium
21	C2.414.7	C2.414	12.93	[1.9, 23.9]	Medium
22	C2.435.104	C2.435	12.85	[1.8, 23.9]	Medium
23	C2.400.74	C2.400	12.65	[1.6, 23.7]	Medium
24	C2.388.61	C2.388	12.63	[1.6, 23.7]	Medium
25	C2.383.99	C2.383	12.62	[1.6, 23.6]	Medium